

SHOPIFY INC.

Monte Carlo Simulation — Data Sources Guide

Quantifying Risk: Production Cost Variations | Demand Fluctuation | Market Entry Risks

Enterprise Risk Management | ALY 6130 | Vantage Discovery Acquisition

Overview

This document provides a structured reference of data sources, distribution parameters, and empirical benchmarks required to build a Monte Carlo simulation for Shopify's risk quantification analysis. The simulation covers three primary risk dimensions: **Production Cost Variations**, **Demand Fluctuation**, and **Market Entry Risks**, all in the context of Shopify's acquisition of Vantage Discovery (March 2025). Each section identifies primary and secondary data sources, recommends probability distribution types, and provides calibrated input parameters derived from Shopify's publicly reported financial data (SEC filings 2024–2025) and industry benchmark reports.

1. Production Cost Variations

Primary Source

Shopify Inc. SEC Filings — Form 6-K and Form 8-K (Q1 2024 – Q2 2025)

These quarterly press releases contain audited income statement data including all operating expense line items. By collecting eight consecutive quarters of data, you can calculate the empirical mean and standard deviation for each cost category and fit appropriate probability distributions for the simulation.

Access at: <https://www.sec.gov> — search entity 'Shopify Inc.' Form 6-K / 8-K

Quarterly Operating Expense Data (USD millions)

Quarter	Revenue	S&M;	R&D;	G&A;	Txn Losses	Total OpEx
Q1 2024	\$1,861	\$361	\$335	\$124	\$51	\$871
Q2 2024	\$2,045	\$353	\$349	\$60	\$42	\$804
Q3 2024	\$2,162	\$331	\$332	\$114	\$58	\$835
Q4 2024	\$2,812	—	—	—	—	—
Q1 2025	\$2,360	\$405	\$377	\$115	—	—
Q2 2025	\$2,844	\$415	\$394	\$122	\$80	\$1,011

Recommended Distribution Parameters

Cost Line	Distribution	Min (\$M)	Most Likely (\$M)	Max (\$M)	Rationale
R&D;	PERT	\$332	\$375	\$415	LLM integration adds upward pressure
Sales & Marketing	PERT	\$331	\$405	\$450	AI launch campaign drives variability
G&A;	Normal	\$114	\$118	\$124	Relatively stable; low variance
Transaction Losses	Lognormal	\$42	\$65	\$80	Right-skewed; large loss events possible

Note. Data sourced from Shopify SEC Form 6-K and 8-K filings, Q1 2024 – Q2 2025. Vantage Discovery integration costs are not separately disclosed; model as an additive range of \$50M–\$150M in R&D; using a uniform distribution for sensitivity analysis.

2. Demand Fluctuation

Primary Source

Shopify Quarterly GMV Reports — SEC Form 6-K / 8-K Press Releases (2022–Q1 2025)

Gross Merchandise Volume (GMV) is Shopify's primary leading demand indicator. Six consecutive quarters of GMV data, combined with the Black Friday/Cyber Monday seasonal spike, provide sufficient data points to fit both a baseline growth distribution and a seasonal peak distribution.

Quarterly GMV Data for Distribution Fitting

Quarter	GMV (\$B)	YoY Growth	Distribution Note
Q1 2024	\$60.9B	23%	Baseline growth range anchor
Q2 2024	\$67.2B	22%	Lower bound of growth range
Q3 2024	\$69.7B	24%	Mid-range; most likely value
Q4 2024 (BFCM)	\$94.5B	26%	Seasonal peak — separate lognormal dist.
Q1 2025	\$74.8B	23%	Post-peak reversion; upper baseline
BFCM 2024 only	\$11.5B	24%	5-day peak event anchor for extreme scenario

Recommended Demand Distribution Parameters

Scenario	Distribution	Min	Base	Max	Source
Baseline quarterly GMV growth	Triangular	20%	24%	30%	Shopify SEC 8-K Q1–Q3 2024
Q4 seasonal demand spike	Lognormal	22%	26%	35%	BFCM press release (SEC 2025)
Merchant churn (demand erosion)	Normal	2%	4%	7%	Recurly 2024 / Riving.io 2026
International GMV growth	PERT	28%	33%	42%	Shopify Q4 2024 investor report

Secondary & Benchmark Sources

- **U.S. Census Bureau — Quarterly Retail E-Commerce Sales:** Market-level demand benchmark for calibrating Shopify GMV growth relative to the total US ecommerce market. Available at census.gov/retail.
- **Recurly 2024 Subscription Benchmarking Report:** Provides industry churn benchmarks by sector (digital media 5.5% monthly, consumer goods 4.1% monthly) for demand erosion scenarios.
- **Shopify BFCM Press Release (SEC Filed Nov 2025):** Documents peak infrastructure demand — 489 million requests per minute — as the anchor for extreme demand scenarios in the simulation.
- **Riving.io Ecommerce Churn Statistics 2026:** Confirms traditional ecommerce annual churn of 70–75%; subscription ecommerce 3.4% monthly. Use for downside demand scenario calibration.

3. Market Entry Risks

Market Share Baseline Data

Market Segment	Shopify Share	Year	Source
US ecommerce market (GMV-based)	>14%	2025	Shopify SEC Form 8-K FY2026
US ecommerce software market	~29%	2025	Statista / BuiltWith via RedStag 2025
Global ecommerce platform market	10.32%	2025	Statista / BuiltWith via RedStag 2025
Global ecommerce software (#1 by sites)	~20%	2025	DemandSage 2026
Active storefronts globally	4.8M stores	2025	RedStag / Chargeflow 2025–2026

Merchant Failure & Attrition Benchmarks

Metric	Rate	Segment	Source
New Shopify store 90-day survival	10%	All new stores	Upsella 2026
Ecommerce first-year failure rate	~70%	All ecommerce	SME News Feb 2025
SaaS annual churn — established	5–7%	Enterprise SaaS	Bold Commerce 2025
SaaS annual churn — new companies	10–15%	Early-stage	Bold Commerce 2025
eCommerce monthly churn — subscriptions	5–10%	Subscription	Recurly 2024
Traditional ecommerce annual churn	70–75%	Non-subscription	Riving.io 2026 / Rivo 2026
Digital media monthly churn	5.5%	Digital media	Recurly 2024 via Upcounting 2025
Consumer goods monthly churn	4.1%	Consumer goods	Recurly 2024 via Upcounting 2025

Monte Carlo Input Parameters — Market Entry Risk

Variable	Distribution	Min	Base	Max	Source
US market share	Normal	12%	14%	16%	Shopify FY2025/2026 SEC filings
Global platform share	Triangular	8%	10.3%	13%	Statista / BuiltWith 2025
New merchant activation growth	PERT	-5%	23%	33%	Shopify GMV / MRR quarterly data
Annual merchant churn rate	Normal	5%	7%	15%	Bold Commerce / Recurly 2024–25
Competitor AI parity timeline	Uniform	6 mo	12 mo	18 mo	Expert judgment / BetaKit 2025
International GMV share	PERT	25%	30%	36%	Shopify Q4 2024 investor report

4. Master Data Source Reference

The table below consolidates all recommended data sources by risk type, data format, and access method for efficient simulation setup.

Risk Category	Source	Data Type	Access
Production Cost	Shopify SEC 8-K / 6-K (Q1 2024 – Q2 2025)	Quarterly OpEx line items (S&M, R&D, G&A, Txn Losses)	sec.gov — EDGAR search 'Shopify Inc.'
Demand Fluctuation	Shopify GMV Press Releases (SEC filed quarterly)	GMV, YoY growth rate, MRR, seasonal BFCM data	sec.gov / investors.shopify.com
Demand Fluctuation	U.S. Census Bureau Quarterly E-Commerce Sales	US retail ecommerce market size (quarterly, seasonally adj.)	census.gov/retail/esales.html
Demand — Churn	Recurly 2024 Subscription Benchmarking Report	Monthly churn rates by industry vertical	recurly.com/resources/benchmarks
Demand — Churn	Riving.io Churn Statistics 2026	67-point benchmark study; annual & monthly churn rates	ringly.io/blog/churn-statistics
Market Entry — Share	Statista / BuiltWith (via RedStag 2025)	Global & US platform market share percentages	statista.com / redstagfulfillment.com
Market Entry — Share	DemandSage Shopify Market Share 2026	Global website counts, Shopify Plus data, CMS share	demandsage.com/shopify-market-share
Market Entry — Failure	Upsella Shopify Churn Rate 2026	90-day survival rate; enterprise vs. SMB retention	upsella.com/blog/shopify-churn-rate-2026
Market Entry — Competition	BetaKit (March 2025) Vantage Discovery article	Qualitative competitive intelligence; AI parity timeline	betakit.com/shopify-acquires-vantage-discovery
Market Entry — Competition	G2 / Gartner Peer Insights Platform Comparisons	Competitor feature ratings; merchant satisfaction scores	g2.com / gartner.com/peer-insights
Peak Demand Scenario	Shopify BFCM 2025 Press Release (SEC)	489M req/min peak; \$11.5B GMV in 5 days	sec.gov Form 8-K Nov 2025 filing

Note. All Shopify financial data sourced from SEC EDGAR (public filings). Churn benchmarks sourced from Recurly and third-party analytics platforms. Market share figures sourced from Statista, BuiltWith, and DemandSage (2025–2026). Qualitative competitive intelligence sourced from BetaKit (2025) and industry analyst reports.

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